Manual Chapter : Load Balancing Passive Mode FTP Traffic with Data Channel Optimization

Applies To:

Show Versions

Configuring a Simple Intranet

Configuring ISP Load Balancing

Routing Based on XML Content

Configuring nPath Routing

Configuring Layer 3 nPath Routing

Creating a Basic Web Site and E-commerce Configuration

Installing a BIG-IP System Without Changing the IP Network

Enabling IP Address Intelligence

Configuring Content Adaptation for HTTP Requests

Configuring Content Adaptation for HTTP Requests and Responses

Configuring HTTP Load Balancing with Source Address Affinity Persistence

Configuring HTTP Load Balancing with Cookie Persistence

Compressing HTTP Responses

Using Via Headers to Acquire Information About Intermediate Routers

Configuring the BIG-IP System as a Reverse Proxy Server

Load Balancing Passive Mode FTP Traffic

Load Balancing Passive Mode FTP Traffic with Data Channel Optimization

Configuring the BIG-IP System as a DHCP Relay Agent

Configuring the BIG-IP System for DHCP Renewal

Configuring a One-IP Network Topology

Configuring the BIG-IP System to Auto-Populate Pools

Implementing Health and Performance Monitoring

Preventing TCP Connection Requests From Being Dropped

Setting Connection Limits

Load Balancing to IPv6 Nodes

Mitigating Denial of Service Attacks

Implementing APM System Authentication

Configuring an SSL Intercept Explicit Proxy Mode

Configuring an Explicit HTTP Proxy Chain

Manipulating HTTPS Traffic by Using a Third-Party Device

Legal Notices

Load Balancing Passive Mode FTP Traffic with Data Channel Optimization

Overview: FTP passive mode load balancing with data channel optimization

You can set up the BIG-IP system to load balance passive mode FTP traffic, with optimization of both the FTP control channel and the data channel.

By default, the BIG-IP system optimizes FTP traffic for the control channel, according to the configuration settings in the default client and server TCP profiles assigned to the virtual server. When you use this particular implementation, you also configure the system to take advantage of those same TCP profile settings for the FTP data channel. This provides useful optimization of the data channel payload.

Task summary for load balancing passive mode FTP traffic

You can perform these tasks to configure FTP passive mode load balancing that optimizes traffic on both the control channel and data channel.

Creating a custom FTP profile

You create a custom FTP profile when you want to fine-tune the way that the BIG-IPsystem manages FTP traffic. This procedure creates an FTP profile and optimizes the way that the BIG-IP system manages traffic for the FTP data channel.

1. On the Main tab, click **Local Traffic** › **Profiles** › **Services** › **FTP** .
   The FTP profile list screen opens.
2. Click **Create.**
   The New FTP Profile screen opens.
3. In the **Name** field, type a unique name for the profile.
4. From the **Parent Profile** list, select the default **ftp** profile.
5. Select the **Custom** check box.
6. For the **Inherit Parent Profile** setting, select the check box.
   This optimizes data channel traffic.
7. Click **Finished.**

The custom FTP profile now appears in the FTP profile list screen.

Creating a custom FTP monitor

An FTP monitor requires a user name and password, and the full path to the file to be downloaded.

The BIG-IP system does not save the downloaded file.

Create a custom FTP monitor to verify passive mode File Transfer Protocol (FTP) traffic. The monitor attempts to download a specified file to the */var/tmp* directory. If the file is retrieved, the verification is successful.

The BIG-IP system does not save the downloaded file.

1. On the Main tab, click **Local Traffic › Monitors** .
   The Monitors List screen opens.
2. Click **Create**.
   The New Monitor screen opens.
3. In the **Name** field, type a name for the monitor.
4. From the **Type** list, select **FTP**.
   The screen refreshes, and displays the configuration options for the **FTP** monitor type.
5. From the **Import Monitor** list, select an existing monitor.

   The new monitor inherits initial configuration values from the existing monitor.
6. In the **Interval** field, type a number that indicates, in seconds, how frequently the system issues the monitor check. The default is `10` seconds.

   The frequency of a monitor check must be greater than the value of the global-level **Heartbeat Interval** setting. Otherwise, the monitor can acquire out-of-date data.
7. In the **Timeout** field, type a number that indicates, in seconds, how much time the target has to respond to the monitor check. The default is `31` seconds.

   If the target responds within the allotted time period, it is considered up. If the target does not respond within the time period, it is considered down.
8. Type a name in the **User Name** field.
9. Type a password in the **Password** field.
10. In the **Path/Filename** field, type the full path and file name of the file that the system attempts to download.

    The health check is successful if the system can download the file.
11. For the **Mode** setting, select one of the following data transfer process (DTP) modes.

Passive	The monitor sends a data transfer request to the FTP server. When the FTP server receives the request, the FTP server initiates and establishes the data connection.
Port	The monitor initiates and establishes the data connection with the FTP server.

12. From the **Configuration** list, select **Advanced**.

    This selection makes it possible for you to modify additional default settings.
13. For the **Up Interval** setting, specify whether to use the up interval:
    - If you do not want to use the up interval, retain the default, **Disabled**.
    - To use the up interval, select **Enabled**, and specify how often you want the system to verify the health of a resource that is up.
14. In the **Time Until Up** field, type a number that indicates the number of seconds to wait after a resource first responds correctly to the monitor before setting the resource to up.

    The default value is `0` (zero), which disables this option.
15. For **Manual Resume**, specify whether the system automatically enables the monitored resource when the monitor check is successful.

    This setting applies only when the monitored resource has failed to respond to a monitor check.

Yes	The system does nothing when the monitor check succeeds, and you must manually enable the monitored resource.

No	The system automatically re-enables the monitored resource after the next successful monitor check.

16. For the **Alias Address** setting, specify an alias IP address:
    - Retain the ***All Addresses** default option.
    - Type an alias IP address for the monitor to verify, on behalf of the pools or pool members with which the monitor is associated.

    If the health check for the alias address is successful, the system marks all associated objects up. If the health check for the alias address is not successful, then the system marks all associated objects down.

17. For the **Alias Service Port** setting, specify an alias port or service for the monitor to check:
    - Accept the ***All Ports** default option.
    - Select an alias port or service for the monitor to check, on behalf of the pools or pool members with which the monitor is associated.

    If the health check for the alias port or service is successful, the system marks all associated objects up. If the health check for the alias port or service is not successful, then the system marks all associated objects down.

18. For the **Debug** setting, specify whether you want the system to collect and publish additional information and error messages for this monitor.

    You can use the log information to help diagnose and troubleshoot unsuccessful health checks. To view the log entries, see the **System › Logs** screens.

Yes	The system redirects error messages and other information to a log file created specifically for this monitor.
No	The system does not collect additional information or error messages related to this monitor. This is the default setting.

19. Click **Finished.**

You can associate the new custom monitor with the pool that contains the FTP resources.

Creating a pool to manage FTP traffic

To load balance passive mode FTP traffic, you create a load balancing pool. When you create the pool, you assign the custom FTP monitor that you created in the previous task.

1. On the Main tab, click **Local Traffic › Pools .**
   The Pool List screen opens.
2. Click **Create.**
   The New Pool screen opens.
3. In the **Name** field, type a unique name for the pool.
4. For the **Health Monitors** setting, in the **Available** list, select a monitor type, and click **<<** to move the monitor to the **Active** list.

   Hold the Shift or Ctrl key to select more than one monitor at a time.

5. From the **Priority Group Activation** list, select **Disabled.**
6. Add each resource that you want to include in the pool using the **New Members** setting:
   a. Type an IP address in the **Address** field.
   b. Type 21 in the **Service Port** field, or select **FTP** from the list.
   c. (Optional) Type a priority number in the **Priority** field.
   d. Click **Add.**

7. Click **Finished.**

The pool to manage FTP traffic appears in the Pools list.

Creating a virtual server for FTP traffic

You can define a virtual server that references the FTP profile and the FTP pool.

1. On the Main tab, click **Local Traffic › Virtual Servers .**
   The Virtual Server List screen opens.
2. Click **Create.**
   The New Virtual Server screen opens.
3. In the **Name** field, type a unique name for the virtual server.
4. For the **Destination Address/Mask** setting, confirm that the **Host** button is selected, and type the IP address in CIDR format.

   The supported format is address/prefix, where the prefix length is in bits. For example, an IPv4 address/prefix is `10.0.0.1` or `10.0.0.0/24`, and an IPv6 address/prefix is `ffe1::0020/64` or `2001:ed8:77b5:2:10:10:100:42/64`. When you use an IPv4 address without specifying a prefix, the BIG-IP system automatically uses a `/32` prefix.

   The IP address you type must be available and not in the loopback network.

5. In the **Service Port** field, type `21` or select **FTP** from the list.
6. From the **FTP Profile** list, select the custom profile that you created earlier.
7. Locate the Resources area of the screen; for the **Related iRules** setting, from the **Available** list, select the name of the iRule that you want to assign and move the name to the **Enabled** list.

   This setting applies to virtual servers that reference a profile for a data channel protocol, such as FTP or RTSP.

8. In the Resources area of the screen, from the **Default Pool** list, select the relevant pool name.
9. Click **Finished.**

The custom FTP virtual server appears in the Virtual Servers list.

Implementation result

A BIG-IP system with this configuration can process FTP traffic in passive mode, in a way that optimizes the traffic on both the control channel and the data channel. This optimization is based on the settings of the default client-side and server-side TCP profiles.

Contact Support

HAVE A QUESTION?

Support and Sales >

FOLLOW US

ABOUT F5

Corporate Information
Newsroom
Investor Relations
Careers
About AskF5

EDUCATION

Training
Certification
F5 University
Free Online Training

F5 SITES

F5.com
DevCentral
Support Portal
Partner Central
F5 Labs

SUPPORT TASKS

Read Support Policies
Create Service Request
Leave feedback [+]

©2023 F5, Inc. All rights reserved.

Trademarks Policies Privacy California Privacy Do Not Sell My Personal Information Cookie Preferences